

Analyzing the Game: Predicting Pro Bowl NFL Quarterbacks Using Performance Metrics

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Introduction

For centuries, sports have been the foundation of the entertainment industry for millions of people all over the world. Sports have united the community and sparked passions throughout generations. In today's world, sports remain deeply cherished by fans, but the games and their analysis have evolved significantly, becoming far more intricate and sophisticated than when they first began. Generally, when family and friends watch sports, it would be to enjoy each other's company and bond over a common interest. Now, it is slightly different: The components indicated are displayed, but now, everything is regarding numbers and data. In the last few decades, sports analytics have played an interesting role in how avid sports fans watch games. As data-driven decisions continue to change the overall of sports, fans, managers, and owners are viewing basketball, football, soccer, and baseball from an entirely new perspective. The use of advanced metrics and analytics has transformed how they evaluate performance, strategy, and player potential. Fans are now more informed and engaged, analyzing statistics and trends alongside traditional game highlights. With the new prevalence of data information, managers and owners can make precise decisions about team building and creating effective strategies. This shift has created a more dynamic relationship between all parties involved in sports.

Additionally, American sports have a friendly exhibition match featuring premier players. Players usually play this game in the middle of the season before the latter half begins to avoid fatigue. This event is called the All-Star Game in basketball (NBA), baseball (MLB), and soccer (MLS). They call the National Football League (NFL) game the Pro Bowl. To pinpoint how these major leagues identify which player fits the criteria can be challenging, so they use multiple factors to select a player. This includes fan voting, impact, marketability, and data statistics. Each component is important and plays a role, but its impact is relatively small. The most crucial factor is statistical data because it objectively demonstrates a player's performance through numbers. 'Numbers do not lie' is the famous idiom often mentioned; Numbers are empirical and provide concrete evidence of ability and contribution. However, over the years, the NFL has had some troubling issues with selecting Pro Bowl quarterbacks (QB), with some players with poor performances, at times, making the selective game because of other factors. Therefore, this paper will serve as a foundation for how NFL fans, coaches, and executives can utilize statistical metrics to identify quarterbacks QB most likely to earn a Pro Bowl selection.

With the basis of descriptive and predictive analytics, focusing strictly on QB statistics, this study aims to establish objective criteria for evaluating QB performance beyond traditional scouting methods.

Predictive analysis can help the NFL maintain a fluid and consistent selection process for quarterbacks in the Pro Bowl. This implementation will help strengthen marketing and fan engagement because it will identify rising stars early. This method will recognize the most deserving quarterbacks in the decision process, assuring the integrity of the league. This study will use a multivariate regression analysis to identify the key performance metrics that contribute to an NFL quarterback's Pro Bowl selection. It will use general statistical indicators such as passing yards, touchdowns, and efficiency ratings, which is what most people typically base their analysts upon, but it will include a few more advanced metrics that will help predict how to access a well-earned Pro Bowl QB selection.

In the National Football League (NFL), quarterback (QB) performance is one of the most heavily analyzed and debated aspects of the game. Each year, Pro Bowl selections are meant to represent the top-performing players at each position. However, these selections are often influenced by subjective factors such as fan voting, media narratives, team success, and popularity rather than purely objective performance metrics. This creates a gap between actual player performance and recognition. From a business analytics perspective, NFL teams invest millions of dollars in quarterback talent, making it critical to have accurate, data-driven evaluations. This project addresses the problem of whether Pro Bowl selections truly reflect quarterback performance. By applying predictive analytics, this study aims to determine if statistical performance alone can accurately predict Pro Bowl selections. This has implications not only for sports analytics but also for decision-making processes in talent evaluation, contract negotiations, and performance benchmarking.

Methodology

The objective of this study is to develop a predictive model that determines whether an NFL quarterback will be selected to the Pro Bowl based on a system and proper performance metrics. We will identify key statistical variables, build a predictive measure, evaluate model accuracy, and compare traditional and advanced metrics to help build the model. This study examines quarterback passing data from 2001 to 2023 obtained from Kaggle. It covers 2,351 observations. The target variable is Pro Bowl selection (1 = selected, 0 = not selected), while the

predictor variables are passing yards, touchdowns, interceptions, completions, passer rating, and Adjusted Net Yards per Attempt (ANY/A).

Before analysis, we cleaned and preprocessed the dataset to improve consistency and reduce unnecessary noise and duplicate data. To handle missing values, we use median imputation. Outliers were treated using the interquartile range method for numerical variables, while the binary target variable was excluded from that process. Aggregate team rows were removed, and sack-related variables, which were defensive measures, were dropped because they were not essential to evaluating quarterback's passing performance. Exploratory analysis was then conducted using histograms, boxplots, scatter plots, correlation heatmaps, and crosstab analysis to examine variable distributions and relationships.

The report treats the Pro Bowl selection as a binary classification task and uses logistic regression as the primary predictive model rather than a general multivariate regression approach. After cleaning the dataset, the data are split into training and testing sets to evaluate unseen observations and filter effectively. The split was 80 training and 20 testing. In doing so, the model performance will help draw an accuracy score, a confusion matrix, precision, recall, and F1-score.

Analysis and Results

The analysis focused on evaluating how well the logistic regression model could predict whether an NFL quarterback would be selected to the Pro Bowl based on performance statistics. The model was trained using variables such as passing yards, touchdowns, interceptions, completion percentage, quarterback rating, and ANY/A. After splitting the data into training and testing sets, the model generated predictions that were evaluated using an accuracy score, confusion matrix, classification report, and coefficient analysis. These evaluation methods helped measure both the overall performance of the model and the importance of each feature in predicting Pro Bowl selections.

The logistic regression model was developed to predict whether an NFL quarterback would be selected to the Pro Bowl based on performance statistics such as passing yards, touchdowns, interceptions, completion percentage, quarterback rating, and ANY/A. After cleaning the dataset and splitting it into training and testing sets, the model was trained and evaluated using multiple performance metrics. The model achieved an accuracy score of approximately 94%, meaning it was able to correctly classify quarterbacks as Pro Bowl

selections or non-selections most of the time. The confusion matrix showed that the model correctly identified a large number of quarterbacks who were not selected while also accurately predicting many quarterbacks who did make the Pro Bowl. The classification report supported these results with precision and recall scores ranging from about 0.80 to 0.86 and an overall F1-score near 0.83, showing that the model performed consistently across both classes.

The model produced strong overall results on the test data. It achieved an accuracy score of **94.61%**, showing that it correctly classified most quarterbacks as either Pro Bowl selections or non-selections. The confusion matrix showed that **418** quarterbacks were correctly identified as not selected and **21** were correctly identified as selected, while **10** were false positives and **15** were false negatives. The classification report showed that the model performed much better on non-selections than selections, with the Pro Bowl class recording **0.68 precision**, **0.58 recall**, and an **F1-score of 0.63**. This suggests that while the model was effective overall, it was still less consistent when identifying quarterbacks who actually made the Pro Bowl

Moreover, the accuracy score showed that the model was able to correctly classify a large portion of quarterbacks in the testing dataset, indicating that player performance metrics do have predictive value when it comes to Pro Bowl selection. The confusion matrix provided a more detailed breakdown of the model's performance by showing the number of correct and incorrect predictions for both selected and non-selected players. The classification report further supported these findings by displaying precision, recall, and F1-scores, which helped evaluate how balanced and reliable the predictions were across both classes.

The coefficient analysis showed that certain variables had a stronger influence on Pro Bowl selection than others. Touchdowns, quarterback rating, and ANY/A appeared to positively impact a quarterback's chances of being selected, while interceptions negatively affected those chances. This suggests that both traditional and advanced quarterback metrics are important in evaluating overall performance. The results also support the idea that more efficient quarterbacks are more likely to receive recognition through Pro Bowl selections.

Although the model produced promising results, there are still limitations that affect its overall accuracy. Pro Bowl selections are not determined strictly by statistics, as factors such as fan voting, team success, media attention, and popularity can also influence outcomes. Because of this, the model cannot fully explain every Pro Bowl selection using performance metrics alone. Future improvements could include adding more variables related to team performance or

experimenting with more advanced machine learning models to improve prediction accuracy and overall model performance.

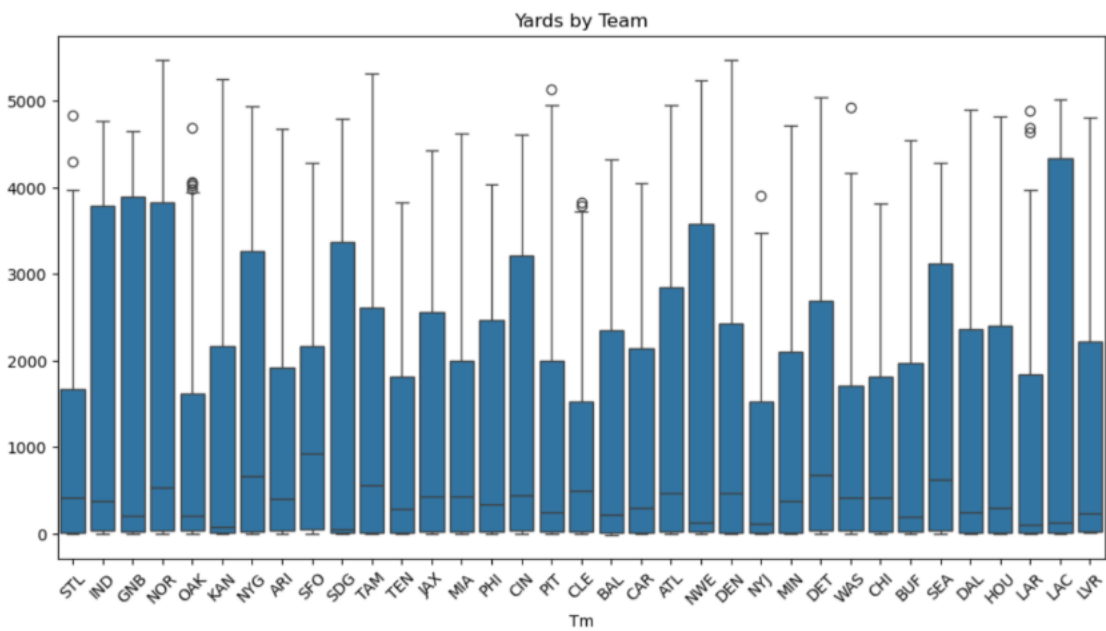


Figure 1

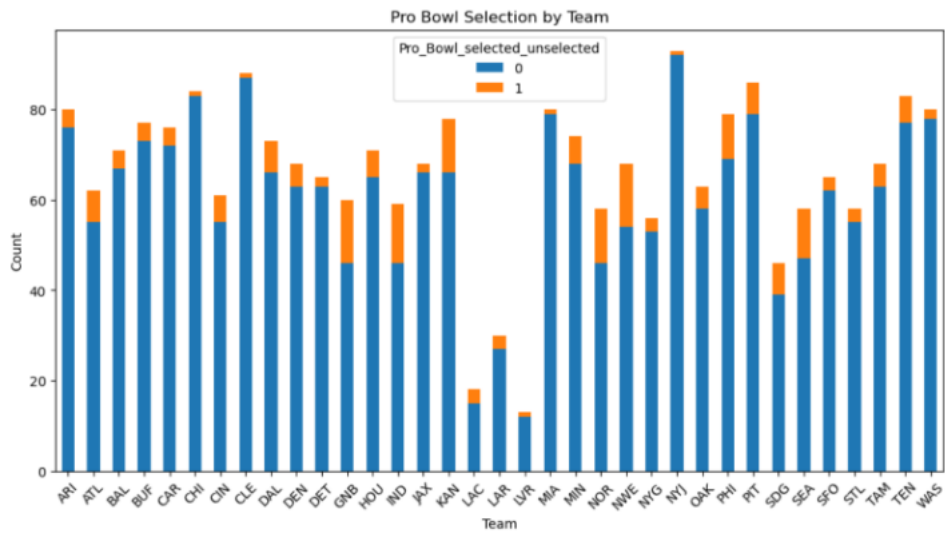


Figure 2